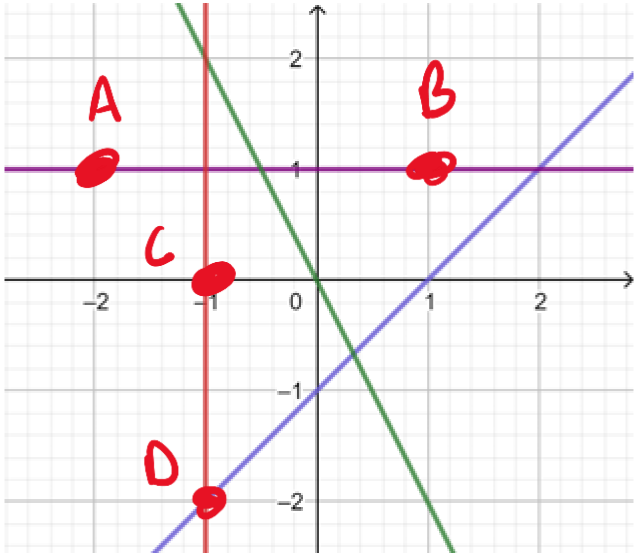
Additional Topic – Differentiation from first principles

Gradients – Rates of Change

Gradients measure rates of change. The gradient of any graph is a measure of the rate at which the *y*-value (the dependant variable) is changing as the *x*-value (independent variable) varies.

For a straight line, the rate of change is constant. The gradient can be calculated anywhere on the line using: $m = \dfrac{\text{rise}}{\text{run}} = \dfrac{y_2 - y_1}{x_2 - x_1}$

-2)



Think:

x_1, y_1 $\quad$ $x_2 y_2$

(a) What is the gradient of the horizontal line?

$A = (-2, 1)$ $\quad$ $B = (1, 1)$

$$m = \frac{y_2 - y_1}{x_2 - x_1} = \frac{1-1}{1-(-2)} = \frac{0}{3} = 0$$

(b) Can you find the gradient of the vertical line?

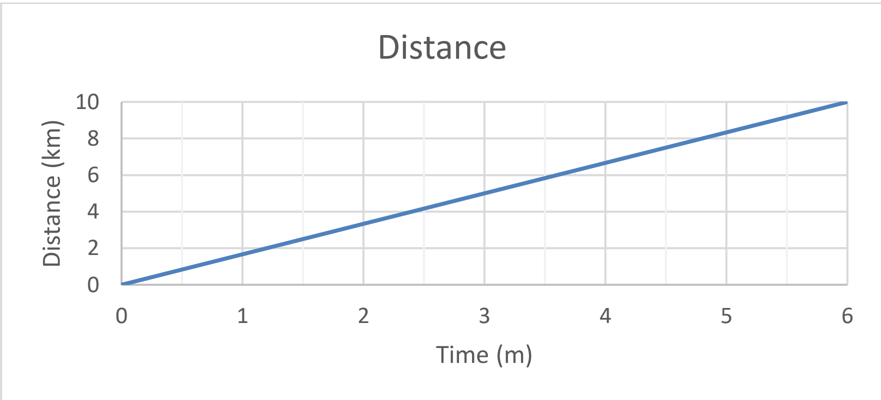
Why/why not?

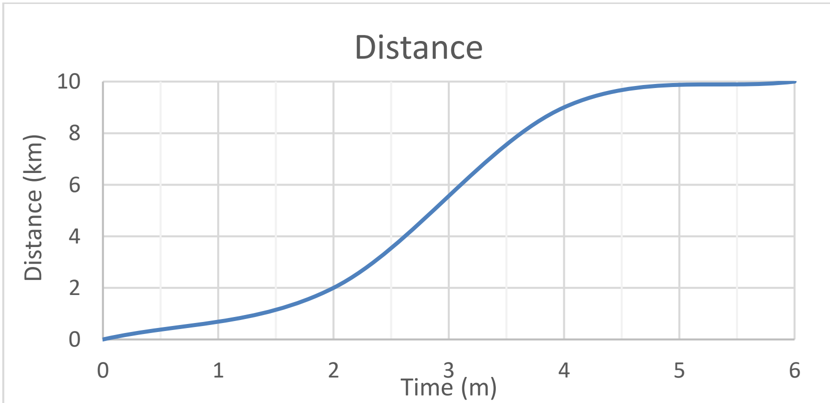
$C = (-1, 0)$ $\quad$ $D = (-1,$

$$m = \frac{-2-0}{-1-(-1)} = \frac{-2}{0} \text{ dne}$$

cannot divide by 0

Investigate: The two graphs below show the distance that a car travels over time.





(a) Is the average speed of the car the same for the two graphs? Average speed $= \dfrac{\text{Total distance (D)}}{\text{Total time (t)}}$

average speed of graph 1: $\frac{10km}{6m}$ average speed of graph 2: 10km/6m

(b) What is different about the speed of the car in the second graph?

The speed is not constant in graph 2 as in graph 1.

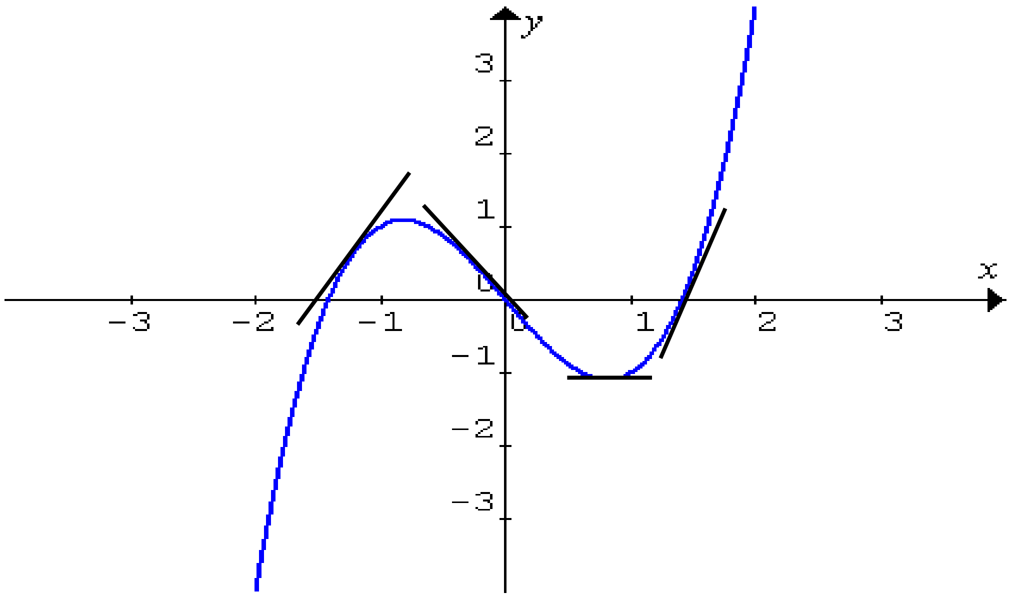
- At the start (0-2) the slope is small → slow speed
- Between (2-4) the slope gets steeper
- Between (4-6) the slope is flattens. → speed decreases

(c) How could you measure the speed of the car in the second graph at a specific point in time?

To measure the speed at a specific point we need to find the gradient of a tangent at a specific point.

Gradient of a Curve

Unlike a straight line, which has a **fixed** or **constant** gradient, the gradient of a curve **changes** continually along the curve (i.e. as x varies).



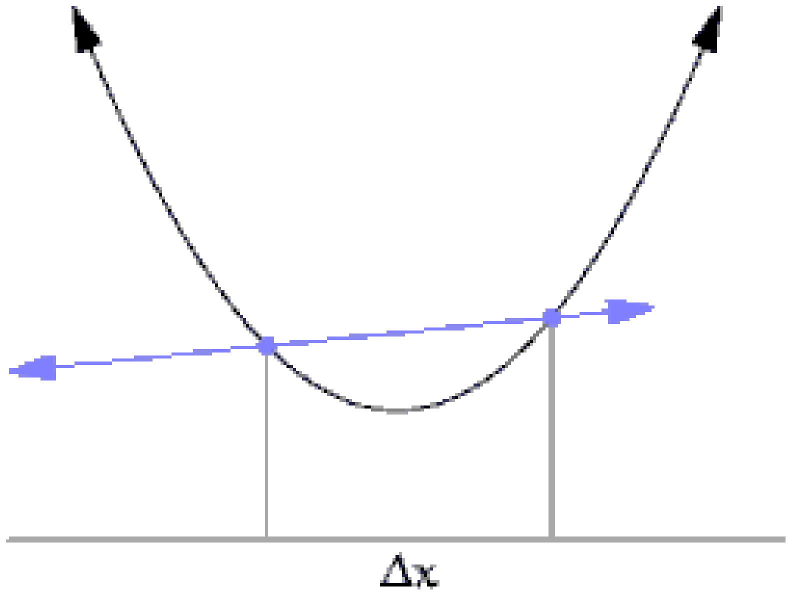
By drawing tangent lines along the graph, we can see where the gradient is positive, negative or zero. Notice that the gradient is zero at points on the curve where the gradient changes from a positive to a negative gradient. These points are often referred to as turning points or stationary points.

Note: *A tangent to a curve is a straight line that touches the curve at a point.*

Estimating the Gradient of a Tangent

The slope of a curve at any point is the same as the **slope of the tangent** that touches the curve at the same point.

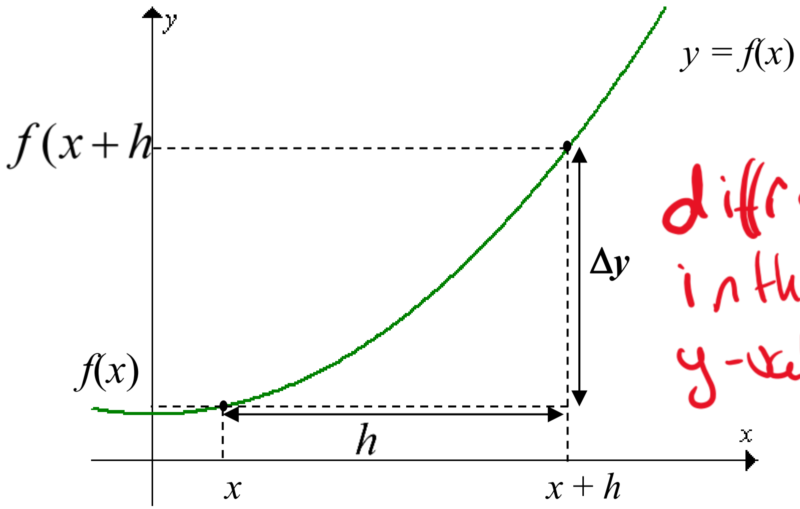
By taking two points that are close together on the graph, the gradient of the tangent line can be approximated by calculating the gradient of the secant (a straight line that cuts the curve at two points). The closer that these two points are together, the better the approximation.



https://commons.wikimedia.org/wiki/File:Tangent_animation.gif#file

Consider the function $y = f(x)$ as the variable x changes from x to $x + h$ where h denotes the increment of the change in x. h is the difference between the last value of x and the first value of x.

As x changes (denoted as Δx) from x to $(x + h)$, y changes from $f(x)$ to $f(x+h)$, which is denoted as Δy.



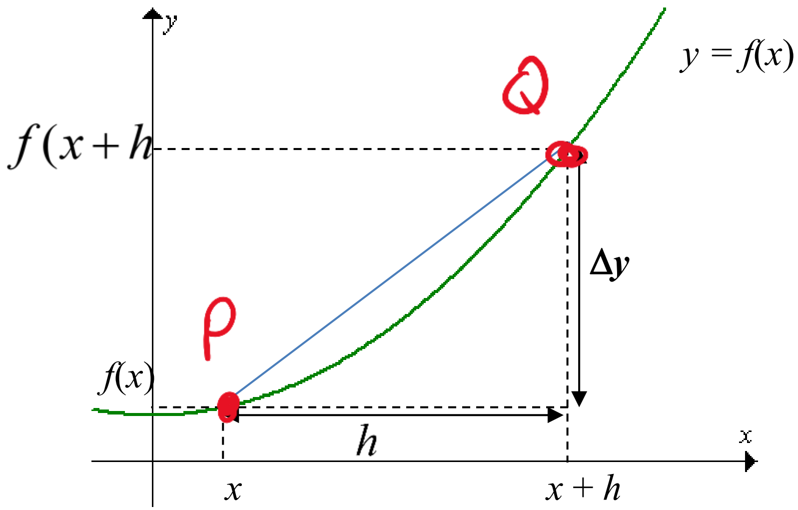
diffrence in the y-value

Now consider the **chord** PQ which is the straight line that joins two points P and Q on the curve.

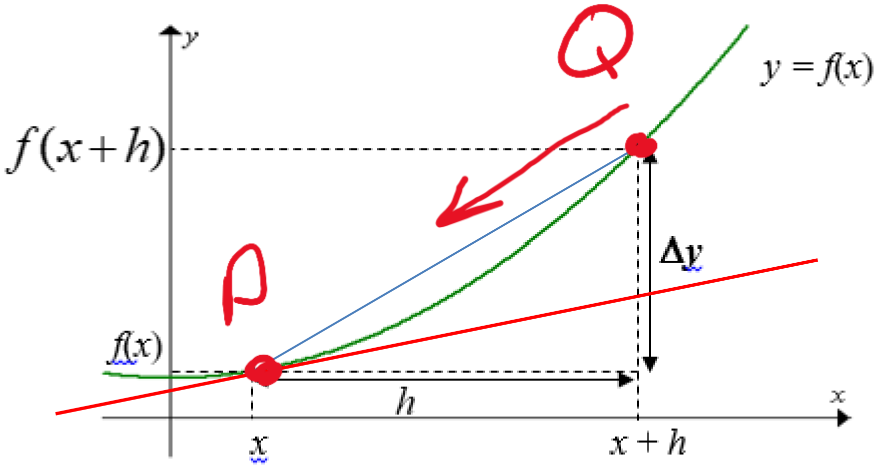
The gradient of the chord gives the **average gradient** of the curve between the two points.

The **average rate of change of $f(x)$** is given by:

$$\frac{\Delta y}{\Delta x} = \frac{y_2 - y_1}{x_2 - x_1} = \frac{f(x+h) - f(x)}{(x+h) - x} = \frac{f(x+h) - f(x)}{h}$$



Now as Q move closer and closer to P.
So h is decreasing. As $Q \to P$, $h \to 0$ and the slope of the secant "approaches" the slope of the tangent line at P.



Therefore, $\dfrac{\Delta y}{\Delta x}$ gives the gradient of the **secant** PQ (or the average rate of change) and

the $\lim\limits_{\Delta x \to 0} \dfrac{\Delta y}{\Delta x}$ gives the gradient of the **tangent** at P (or the **instantaneous rate of change** at P).

Finding the instantaneous rate of change of a curve is also called the **derivative** and it involves taking the **limit** of the *average rate* of change.

The **average rate of change of $f(x)$** is given by:
$$\frac{\Delta y}{\Delta x} = \frac{f(x+h) - f(x)}{h}$$

Whereas **the gradient of the tangent is** given by:
$$\lim_{h \to 0} \frac{f(x+h) - f(x)}{h}$$

Differentiation from "first principles"

Differentiation from "first principles" involves using algebra to find the gradient function that defines the slope of a curve.

For the function $y = f(x)$, the **derivative of *y* with respect to *x*** is defined by:

$$\frac{dy}{dx} = \lim_{\Delta x \to 0} \frac{\Delta y}{\Delta x} = \lim_{h \to 0} \frac{f(x+h) - f(x)}{h}$$

For convenience $\lim_{\Delta x \to 0} \frac{\Delta y}{\Delta x}$ is denoted by $\frac{dy}{dx}$ or $f'(x)$ or $y'(x)$.

If $y = f(x)$ is a differentiable function, we use the notation $\frac{dy}{dx}$ or $f'(x)$ or $y'(x)$ to refer to the **derivative** or **gradient function** of $y = f(x)$.

***Note:** Some functions are not differentiable.*

Review of Function Notation

In preparation for differentiation, it is important to revise function notation.

Exercise:

If $f(x)=3x^2+x$, find.

(a) $f(2)$ $= 3\times 2^2+2$

$= 3\times 4+2$

$= 12+2$

$= 14$

(c) $f(x+2)$ $= 3(x+2)^2+(x+2)$

$3(x^2+4x+4)+(x+2)$

$3x^2+12x+12+x+2$

$= 3x^2+13x+14$

(b) $f(2)-f(1)$

$= 14-(3\times 1^2+1)$

$= 14-(3\times 1)$

$= 14-4$

$= 0$

(d) $f(x+2)-f(x)$

$=(3x^2+13x+14)-(3x^2+x)$

$= \cancel{3x^2}+13x+14-\cancel{3x^2}-x$

$= 12x+14$

Determining the derivative (gradient function) from "first principles"

When we use limits to determine the derivative that describes the gradient of a curve at any point, this is known as differentiation from first principles. Determining the gradient function using the concept of limits requires careful setting out and strong algebra skills.

$f'(x) = \lim\limits_{\Delta x \to 0} \dfrac{\Delta y}{\Delta x}$	**or**	$f'(x) = \lim\limits_{h \to 0} \dfrac{f(x+h) - f(x)}{h}$

To differentiate a function from first principles:

1. Determine $f(x+h)$
2. Determine $f(x+h) - f(x)$
3. Determine $\dfrac{f(x+h) - f(x)}{h}$
4. Write definition $f'(x) = \lim\limits_{h \to 0} \dfrac{f(x+h) - f(x)}{h}$
5. Lastly let $h = 0$ to determine the limit.

It is important to use the correct notation at each step of differentiation by first principles.

***Example*:**

Use the definition $f'(x)=\lim_{h\to 0}\dfrac{f(x+h)-f(x)}{h}$ to find the gradient function of $f(x)=x^2+1$.

= derivative

Solution:

Step 1:
$$f(x+h)=(x+h)^2+1$$
$$=x^2+2xh+h^2+1$$

Step 2:
$$f(x+h)-f(x)=\left(x^2+2xh+h^2+1\right)-\left(x^2+1\right)$$
$$=x^2+2xh+h^2+1-x^2-1$$
$$=2xh+h^2$$

Step 3:
$$\frac{f(x+h)-f(x)}{h}=\frac{2xh+h^2}{h}=\frac{h(2x+h)}{h}$$
$$=2x+h$$

Step 4 and 5:
$$f'(x)=\lim_{h\to 0}\frac{f(x+h)-f(x)}{h}$$
$$=\lim_{h\to 0}2x+h$$
$$=2x$$

So, the gradient function is $f'(x)=2x$

Exercise:

Find the gradient function of $f(x) = x^2 + 5x$ using first principles $f'(x) = \lim_{h \to 0} \frac{f(x+h) - f(x)}{h}$

- $f(x+h) = (x+h)^2 + 5(x+h)$
  $= x^2 + 2xh + h^2 + 5x + 5h$
- $f(x+h) - f(x) = x^2 + 2xh + h^2 + 5x + 5h - (x^2 + 5x)$
  $= \cancel{x^2} + 2xh + h^2 + \cancel{5x} + 5h - \cancel{x^2} - \cancel{5x}$
  $= 2xh + h^2 + 5h$
- $\frac{f(x+h) - f(x)}{h} = \frac{2xh + h^2 + 5h}{h} = \frac{\cancel{h}(2x + h + 5)}{\cancel{h}} = 2x + h + 5$
- $f'(x) = \lim_{h \to 0} (2x + h + 5)$
  $= 2x + 0 + 5$
  $f'(x) = 2x + 5$

***Example*:** derivative

In a previous example the gradient function of $f(x) = x^2 + 1$. was determined to be $f'(x) = 2x$.

Now find the gradient of tangent to the curve at $x = 1$ and $x = 2$.

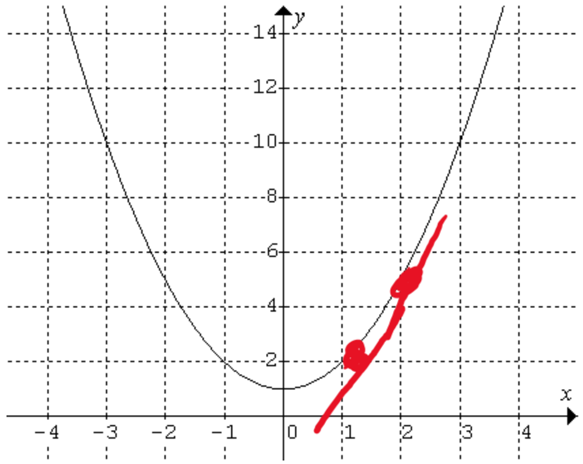
Solution:

This question requires the evaluation of $f'(1)$ and $f'(2)$

$f'(x) = 2x$.

$f'(1) = 2 \times 1 = 2$

$f'(2) = 2 \times 2 = 4$



Geometrically we can see that the gradient of tangent at $x = 1$ is 2 and the gradient of a tangent at $x = 2$ is 4.

The slope of the tangent at **any** point on the curve $f(x) = x^2 + 1$ can be calculated from the gradient function.

Exercise:

In a previous exercise the gradient function of $f(x) = x^2 + 5x$ was determined to be $f'(x) = 2x + 5$.

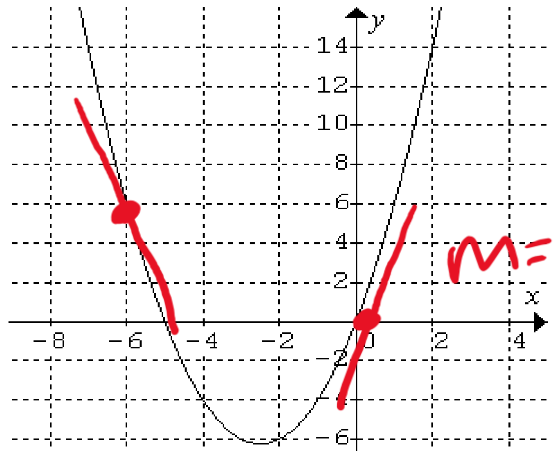
Find the gradient of tangent to the curve at $x = -6$ and $x = 0$.

at $x = -6$ $\quad f'(-6) = 2 \times (-6) + 5 = -7$

at $x = 0$ $\quad f'(0) = 2 \times 0 + 5 = 5$

Confirm that the gradients calculated above are consistent with the graph of $f(x) = x^2 + 5x$

$m = 7$



$m = 5$

Class Questions:

First Principles

1. Using first principles, find the derivative of:

   (i) $5x+2$ (ii) $3x^2+1$

Solutions to Class Questions:

1. (i) $5x+2$

$$f(x)=5x+2$$

$$f(x+h)=5(x+h)+2$$

$$=5x+5h+2$$

$$f(x+h)-f(x)=5x+5h+2-5x-2$$

$$=5h$$

$$\frac{f(x+h)-f(x)}{h}=\frac{5h}{h}$$

$$=5$$

$$f'(x)=5$$

(ii) $3x^2+1$

$$f(x)=3x^2+1$$

$$f(x+h)=3(x+h)^2+1$$

$$=3x^2+6xh+3h^2+1$$

$$f(x+h)-f(x)=3x^2+6xh+3h^2+1-3x^2-1$$

$$=6xh+3h^2$$

$$\frac{f(x+h)-f(x)}{h}=\frac{6xh+3h^2}{h}$$

$$=6x+3h$$